EVOLUTIONARY DISCOVERY OF REINFORCEMENT LEARNING ALGORITHMS VIA LARGE LANGUAGE MODELS

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WHAT RL IS USED FOR

Reasoning LLMs

o1, DeepSeek-R1 — RL makes them think

LLM alignment

RLHF, DPO, GRPO, Constitutional AI

Robotics & autonomy

π_0, OpenVLA, Atlas, Waymo, AlphaStar

Science & operations

AlphaDev, AlphaTensor, chip design, data-center cooling

This paper targets control RL — Gymnasium / MuJoCo; not LLM-RL.

RL VS. ES AT A GLANCE

Reinforcement Learning

Learn a policy by trial & error, updated with gradients on a scalar reward.

Where it runs today

- AlphaGo, AlphaStar, MuZero
- ChatGPT / Claude / Gemini (RLHF)
- o1, DeepSeek-R1 reasoning
- Humanoid locomotion, manipulation

The dominant paradigm.

Evolutionary Search

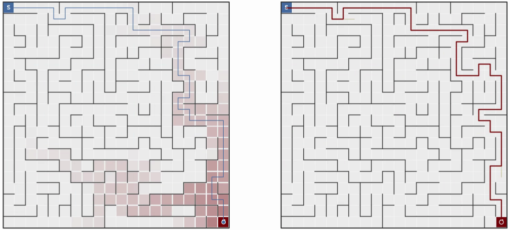
Maintain a population, mutate & recombine, keep the fittest. No gradients.

Where it runs today

- Neural architecture search (NAS)
- Hyperparameter tuning (PBT, PB2)
- Reward discovery (Eureka, REvolve)
- Non-differentiable problems (Fun-Search)

The niche tool — used where RL can't go.

DEMO — WHEN RL WINS

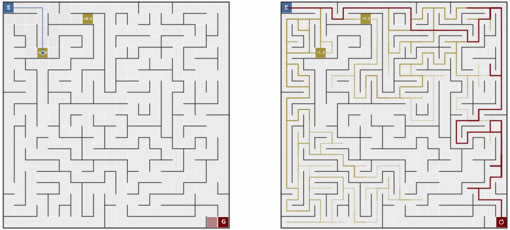


Same 20×20 maze, sparse goal reward.

- RL: 212K env steps, $\sim$50-step path
- ES: 24$\times$ more env steps for the same result

Clean credit assignment is RL's home turf.

DEMO — WHEN ES WINS



Add two small “trap” rewards near the start.

- RL: 0 / 2,000 episodes reach the big goal
- ES (MAP-Elites): 248 / 1,000 generations do
- 302 total breakthroughs · first at gen 44

Deceptive landscapes are where population search wins.

WHAT THIS PAPER IS

- Each candidate is an executable Python `compute_loss` function
- An LLM proposes mutations and crossovers over that code
- Each variant is scored by actually training agents in Gymnasium
- Extends REvolve — aimed at the loss function, not the reward
- Prompt bans actor–critic, value functions, and Bellman updates
- Discovers two new algorithms — CG-FPD and DF-CWP-CP
- Competitive with PPO and SAC across 10 Gymnasium benchmarks

THE CANONICAL RL ALGORITHMS

- DQN (2015) — value-based, discrete action spaces
- TRPO (2015) — trust-region policy gradient
- DDPG (2015) — off-policy continuous control
- PPO (2017) — clipped policy gradient, the workhorse
- SAC (2018) — entropy-regularized off-policy
- TD3 (2018) — twin-critic DDPG

WHAT AUTOML HAS AUTOMATED

- Architecture — automatically picking network shape, depth, width
  NAS, AmoebaNet, AutoML-Zero
- Hyperparameters — learning rate, batch size, clipping, γ
  PB2, OptFormer, PBT
- Reward function — writing the scalar reward the agent optimizes
  Eureka, REvolve
- Preference loss — the loss used to learn from pairwise preference data
  Christiano RLHF, DPO, DiscoPOP, PEBBLE
- The update rule itself — the full RL training loop as code
  open until this paper

TRAINING AS A THREE-TIER FILTER

LLM proposes
∼80 candidate update rules

Tier 1 — smoke test
1 seed, 1 env, short rollout

top ∼20% pass

Tier 2 — focused eval
3 seeds, 3 envs, medium rollout

top ∼20% pass

Tier 3 — full suite
3 seeds, all 5 envs, full training

Cheap filter first.
Expensive analysis only on survivors.
Parallels thinking-mode compute allocation in LLMs.

THE TWO ALGORITHMS IT INVENTED

CG-FPD

Confidence-Guided Forward Policy Distillation

Distill short-horizon plans from a learned latent dynamics model. Planning acts as the supervised teacher signal.

No value function. No policy gradient. No Bellman.

DF-CWP-CP

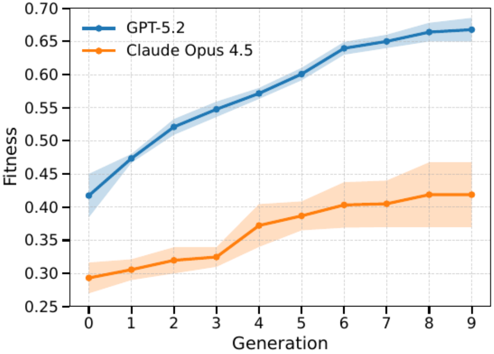
Differentiable Forward Confidence-Weighted Planning with Controllability Prior

Differentiable world-model rollouts + confidence-weighted desirability + controllability prior.

Also avoids every canonical RL mechanism.

Evolution rediscovered model-based planning — from scratch.

THE PROPOSER LLM MATTERS. A LOT.



- GPT-5.2 climbs to 0.67
- Claude 4.5 Opus caps at 0.42
- Both monotone; GPT-5.2 is sharper on every generation

Same framework, same envs, same seeds — just swap the LLM.

RESULTS MATCH PPO AND SAC

Environment	PPO	SAC	CG-FPD	DF-CWP-CP
CartPole	500.0	—	500.0	500.0
LunarLander	246.6	—	241.2	260.6
MountainCar	-128.1	—	-105.8	-108.6
Acrobot	-63.5	—	-90.6	-78.6
HalfCheetah	1579.1	4988.5	2407.9	2103.8
Swimmer	95.0	87.8	247.5	219.8
Walker2d	3163.2	4595.3	1603.8	1297.6
InvertedPendulum	1000.0	1000.0	1000.0	1000.0

Competitive with hand-designed baselines across 10 environments.

Evolved on 5 training envs; remaining 5 are generalization tests.

STRENGTHS VS. WEAKNESSES

Strengths

- First full update-rule discovery via LLM
- Genuinely novel mechanisms (no value, no PG)
- Clean extension of REvolve's machinery
- LLM-HPO handles fitness fragility cleanly

Weaknesses

- 30 h per candidate on $4\times$A100
- Narrow benchmarks (CartPole / MuJoCo only)
- Is GPT-5.2 discovering or remembering?
- Only 2 evolutionary seeds per LLM
- No head-to-head vs. DiscoRL
- Constitutional constraints never ablated

THANK YOU!

Questions & Discussion

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